Review Sheet 4: Some More Bets[1]

Suggested Answers

1. Banking Crisis

You are in the unfortunate but well-paid position of being Chief Investment Officer at a leading hedge fund. While your fund has been performing well, the contagion and fear in capital markets is getting to your investors. This is leading to huge redemptions from your fund. You must plan to meet these redemptions at the lowest costs. You and your economic team employ the Quaker method of estimating these redemptions and predict that over the course of the next month, the expected redemptions will be US$ 300 Million. The standard deviation of this estimate is US$100 million. Unfortunately, all your hedge fund assets are illiquid and cannot be disposed for the next year. You must thus go to the capital markets for raising additional funds.
You have two sources of financing that you are planning to use simultaneously:
Source 1: You can borrow from the government sponsored TARP (The-Augmentation-of-Rich-People) program. This facility charges a usurious 25% annual rate of interest. Further, it can be availed only if a request is made far in advance.
Source 2: Alternately, when TARP funding comes short, you can borrow from the overnight credit markets. You can use this facility anytime and it provides funding immediately. Unfortunately, the rate of interest charged is 40% per annum.

*a) How much money would you borrow from the TARP funds?****

Solution: This is a classic newsvendor setup with reactive capacity: The cost of borrowing too little is Cu = 15, whereas the cost of borrowing too much is Co = 25. Note that units don't matter here.

Critical ratio is 15/(15+25) = 3/8=0.375. From the table, z-value is -0.31 (remember, if we can't find the exact value, we look for the next highest value). So the amount borrowed is

$Q^* = \mu + z^* \times \sigma = 300 - 0.31 \times 100 =$ *US$ 269 million.*

*b) What are your annualized capital costs if you employ the solution prescribed in part A?***

*For 269 million you have to pay 25%, that is 269*0.25 = 67.25 mill. On the other hand, because you need to satisfy whatever demand there is, you will have to borrow from the overnight market if there is demand and you don't have the money available. In other words, if anything above 269 mill is required, you have to borrow. This means that you have to borrow the expected lost sales from the overnight market.*

For z of -0.31, from the table you read: L(z*)= 0.57296. Expected lost sales are then: 100*0.57296= 57.30 mill. For this amount you need to pay 40%, or 22.92 mill.*

[1] Problems are rated for difficulty: * = Relatively easy, for first review of concepts; ** = Challenging, assumes good understanding of the concepts; *** = Advanced, assumes full understanding of the concepts and creativity in application.

Total costs 67.25+22.92=90.17 million.

Note: It is not enough to write down the expected lost sales! It must be clear from the logic of the solution why we need to compute this, i.e. this is the money that you need to borrow from the overnight market.

c) ***One of your classmates at INSEAD is in exactly the same financial position as you. While he runs an identical fund, most of his clients are located in a different geography. He calls you up and proposes merging your two funds. He somewhat remembers the POM class and suggests that merging would reduce the above mentioned capital costs for both funds. Do you agree? Justify your answer in no more than 100 words. *****

Yes, merging the two funds will decrease capital costs for both as you will now aggregate demand for redemptions from both markets. This will have lower uncertainty and this will lead to the capital costs being lower. This is the same principle of aggregating demand from multiple markets that we have seen multiple times in class.

d) ***The crisis is getting worse. The overnight credit markets have now completely frozen up. You no longer care about minimizing your capital costs. All you care about is the odds of your survival, or the probability that your fund will not be able to meet redemptions. What is the minimum amount of money you need to borrow from TARP funds to ensure that you will be able to meet all redemption requests with a 95% probability?******

This translates to a Q such that Pr(Q >= D)=0.95. We can rewrite this as

Pr ((Q-300) /100 >= (D-300)/100)=0.95. Because demand is normal with mean 300 and standard deviation 100, then (D-300)/100 is standard normal. So the above probability is equal to: Pr(Z <= z) = 0.95 with z=(Q-300)/100. From the table the value of z corresponding to 0.95 is 1.65 and therefore: Q-300/100 = 1.65 or Q=465 mill.

2. Energy Economics

A massive natural gas field has been discovered in the beautiful Central Asian Republic of Gasistan. The gas field provides a limitless supply of gas that has the potential to transform the lives of the people (and rulers) of Gasistan. However, to realize this prosperity the National Gas Corporation of Gasistan (NGCG) must develop facilities to transport the plentiful natural gas to energy-hungry markets. The geological makeup of this field dictates that gas that is produced must be transported to markets or it is lost. Assume that for the time-frame of the analysis, the field can produce any quantity of gas that can be transported.
You have been hired as a consultant to the NGCG. Based on your research, you have outlined two potential ways to transport gas to somewhat differing markets.

1. **Pipeline:** *A pipeline can be built to transport a fixed quantity of gas to European markets. Daily demand in the European markets that can be accessed by the proposed pipeline is distributed Normally with a mean of 600 Million Cubic Feet (mcf) and a standard deviation of 600 mcf. The low operation costs of the pipeline ensure that every mcf of gas that can be sold in European markets, NGCG earns a profit of $10,000 (taking into account* ***operating costs****, i.e., production, transportation and all other operating costs). Building a pipeline involves a heavy capital investment. Based on the availability of capital and the sovereign guarantees available to the NGCG, you estimate that the daily* **capital cost** *of one mcf of transportation capacity will be $5,000. In other words, if a pipeline with daily capacity of 1,000 mcf is built, the daily capital cost would be $5 million.*

2. **Liquefying and shipping terminal:** *Natural gas may also be liquefied and can then be sent via special sea-going tankers to a variety of global markets. Daily Demand in the markets that can economically be accessed by sea-going vessels is distributed normally with the same mean as daily demand in European markets, 600mcf. However, the diversity of markets that can be accessed by sea-going vessels limits the impacts of region-specific demand shocks and the standard deviation of the accessible markets is only 200 mcf. Further, you estimate that every mcf of gas that can be sold in these markets will earn a profit of $9,000. Finally, you estimate that the daily capital cost of one mcf of liquefying and transportation capacity will be $6,000.*

a) If you decide to build a pipeline, what is the capacity of the pipeline that maximizes the expected profits?*

Suggested solution: NCGC is facing uncertain demand from the market and has to make a bet on this potential demand, by installing pipeline capacity. A huge pipeline is likely to go unused while with a small pipeline NCGC is likely to lose out on potential sales. The newsvendor model applies in this setting.

To obtain the optimal capacity, we need to calculate the underage and overage costs.

Cu = 10K – 5K = 5K.

10K is the revenue and 5K is the cost of installing one unit.

Co = 5K.

Thus the critical factor is Cu/(Cu+Co)=0.5. This means that in this case NCGC needs to build a capacity of 600mcf. (Note that z=0).*

b) If you decide to build a pipeline with the capacity calculated in A., what would be the average utilization of that pipeline? *

Suggested solution: Utilization is the fraction of the installed capacity gets used (on average). To compute this number, we need the installed capacity and the average capacity used. We already know the total capacity from A. We need to next compute the average capacity used. Average capacity used corresponds to the average sales in a standard newsvendor setup. So we need to compute the expected sales:

*E[Sales]= E[Demand] –E[Lost Sales]= 600-600*L(0)=600-600*0.39894=360.64.*

Utilization = 360.64 / 600 = 60.11%.

c) With the pipeline option, what are the daily profits? (taking into account the capital costs)*

Suggested solution:

*Profits = Sales*10K- Capacity*5K = 360.64*10,000-600*5,000 = $606,360.*

We might take also the following approach (discussed in tutorials):

Expected leftover units (expected unused capacity) = Q - Expected sales = 600 – 360.64 = 239.36*

Expected mismatch cost = Cu × Expected lost sales – Co × Expected leftover units = 5K × 239.36 + 5K × 239.36 = 2393.64 K

Expected Profits = 5K × 600 - 2393.64 K= 606.36 K = $606,360.

d) ***If you decide to build the liquefying facility, what is the profit maximizing capacity of the liquefying and shipping terminal? ****

Suggested solution: We need to repeat the same procedure as for pipeline option.

It is easy to see that Cu= 3000 and Co= 6000 and thus the critical fractile is: 1/3. From Table 1, we have: z= - 0.43 (note this is negative).*

Therefore: Q = μ + z* × σ = 600 – 0.43 × 200 = 514mcf*

e) ***If you decide to build the liquefying facility with the capacity calculated in D, what would be the utilization of that facility?****

Suggested solution:

*E [Sales] =E [Demand]-E[Lost Sales]=600-200*L(-0.43)=600-200*0.65027=469.95. The*

Utilization = 469.95 / 514 =91.43%.

f) ***With the liquefying option, what are the daily profits? (taking into account the capital costs)****

Suggested solution:

*Profits = Sales*9K- Capacity*6K = 469.95*9,000- 514*6,000 = $1,145,514.*

g) ***Based on your computation of daily profits, which of the two options would you chose. Qualitatively, highlight the key difference between the two options that drives your recommendation. ****

Suggested solution: Go for the Liquefying option, because liquefying allows for aggregation of demand from multiple markets. Demand aggregated from multiple markets has lower variance than demand from one market. From the Zara case and the associated class analysis, we know that if demand variance (or uncertainty) is lower, firm profits are higher. Here we see that this can be the case even if the per-unit profit is substantially lower.

3. Land's End

Geoff Gullo owns a small firm that manufactures "Gullo Sunglasses". He has the opportunity to sell a particular seasonal model to Land's End. Geoff offers Land's End two purchasing options:
Option 1: Geoff offers to set his price at $65 and agrees to credit Land's End $53 for each unit Land's End returns to Geoff at the end of the season (because these units did not sell). Since styles change each year, there is essentially no value in the returned merchandise.
Option 2: Geoff offers a price of $55 for each unit, but returns are no longer accepted. In this case, Land's End throws out unsold units at the end of the season.
The season's demand for this model is Normal with mean 200 and standard deviation 125. Land's end will sell those sunglasses for $100 each. Geoff's production cost is $25.

a) ***How much would Land's End buy if they chose Option 1?****

Solution: With option 1 Land's End sales price is $100, purchase cost is $65 and salvage value is $53 (because Geoff buys back unsold glasses for $53). So the underage cost is C_u *=100 – 65 =35 and the overage cost is* $C_o = 65 - 53 = 12$. *The critical ratio is* $C_u/(C_o + C_u)$ *= 35/47 = 0.7447. From the Standard Normal Distribution Function Table we choose* $z = 0.66$. *The optimal order quantity is then* $Q = \mu + \sigma \times z = 200 + 0.66 \times 125 = 283$ *(note that we have to round up as Land's End cannot order half sunglasses).*

b) How much would Land's End buy is they chose Option 2?*

With option 2 Land's End sales price is $100, purchase cost is $55 and the salvage value is $0. So the underage cost is $C_u = 100 - 55 = 45$ *and the overage cost is* $C_o = 55$. *The critical ratio is* $C_u/(C_o + C_u)$ *= 45/100 = 0.4500. From the Standard Normal Distribution Function Table choose* $z = -0.12$. *The optimal order quantity is then* $Q = \mu + z \times \sigma = 200 - 0.12 \times 125 = 185$.

c) Which option will Land's End choose?*

We need to evaluate expected profits in each case to determine which option Lands End should choose. The process to calculate the expected profits is mechanical, as follows:

Expected lost sales: $125 \times L(0.66) = 125 \times 0.15281 = 19.10$

Expected sales: mean demand - expected lost sales = 200 – 19.10 = 180.90

Expected leftover inventory= Q - expected sales= 283 – 180.90 = 102.10*

*Mismatch cost= 19.10 * 35 + 102.10 * 12 = 1893.76.*

Expected profit= (r-c) mean demand – mismatch cost= 35*200 – 1893.76= 5106.24*

We now need to repeat the same process for option 2. This results is an expected profit of 4052.38. Therefore, option 1 should be chosen.

d) Suppose Land's End chooses option 1 and orders 275 units. What is Geoff Gullo's expected profit?*

REMARK 1: This question asks for GEOFF GULLO's profits and not Land's End profits. Points have been cut for ignoring this.

REMARK 2: This question says that Land's End orders 275 units and not the optimal order quantity calculated before. All the calculations must be based on the new order quantity and not the optimal one. Points have been cut for ignoring this.

Geoff first needs to produce 275 which cost 25 each. Then, it sells them to Land's End at 65 each. But it also needs to take back the unsold ones (or expected leftover inventory for the Land's End).

The only component we don't have is the expected leftover inventory, which we calculate as follows:

First find the z-value corresponding to new order quantity: z = (275 – 200) / 125 = 0.6.

Then we have:

*Expected lost sales = 125 * L(z)= 125 * 0.16867 = 21.09*

Expected sales = 200 – 21.09 = 178.92

Expected leftover inventory = 275 – 178.91 = 96.08

Therefore:

*Geoff's Profit: 275*65 – 275*25 – 96.09 * 53 = 5907.56*